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DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

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Energy Conversion II Laboratory

Project Report

Section: C1 Group: 01 & 04

Three-phase Synchronous Generator Design

Course Instructors:

Asikur Rahman Jowel, Lecturer
Shafin-Bin-Hamid, Lecturer

Signature of Instructor: _____

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Group 01	Group 04
Signature: _____ **Full Name: Md. Refat Khan** **Student ID: 2006131**	**Signature: _____** **Full Name: Sad Sami** **Student ID: 2006150**
Signature: _____ **Full Name: Raisa Nusrat** **Student ID: 2006132**	**Signature: _____** **Full Name: Niloy Kumar Sarkar** **Student ID: 2006151**
Signature: _____ **Full Name: Md. Symria Raihan** **Student ID: 2006133**	**Signature: _____** **Full Name: Mst. Sadia Israt Oishe** **Student ID: 2006152**
Signature: _____ **Full Name: S. M. Razibur Rahman** **Student ID: 2006134**	**Signature: _____** **Full Name: Riajul Karim Chowdhury** **Student ID: 2006153**
Signature: _____ **Full Name: Himel Saha** **Student ID: 2006135**	**Signature: _____** **Full Name: Meharab Hossen Bappy** **Student ID: 2006154**
Signature: _____ **Full Name: Bashir Shahriyer Rafi** **Student ID: 2006136**	**Signature: _____** **Full Name: Nawshin Nawar Nijhum** **Student ID: 2006155**
	Signature: _____ **Full Name: Chowdhury Blossom I. Razzaque** **Student ID: 2006156**

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1 Abstract

Our project is to convert a 3phase 4 pole induction motor into a synchronous motor/generator. Induction machines operate at slip speed meaning they can never reach synchronous speed and their speeds seem to drupe with increasing load. Synchronous machines on the other hand operate at constant synchronous speed regardless of loading which is why they are used in constant speed application. We aim to devise a simplistic method to convert the readily available and cheap induction motor into synchronous motor/generator for both power generation purposes and constant speed applications. Besides, synchronous motors can also be used as capacitors (known as synchronous condensers) that help in power factor correction which reduces the cost of installing a power factor improvement plant. Besides, synchronous motors have higher efficiency and no slip when compared with induction motors of similar size. Synchronous generators also offer an easygoing way of generating electrical power and can be easily paralleled with the infinite bus. Synchronous generators can efficiently be converted into induction motors and vice versa though the same can't be said about induction motors. Overall, our aim was to gain knowledge about the construction of both induction motors and synchronous generators and develop an easy way of converting one into the other.

2 Introduction

This project is about constructing a three-phase synchronous generator from a three-phase four-pole induction motor.

Induction motors and synchronous generators are two distinct electrical machines with different operating principles and applications. The main difference between an induction motor and a synchronous generator lies in their operating principles, modes of operation and synchronicity with the electrical supply, applications, and control capabilities.

Induction motors are primarily used for converting electrical energy into mechanical work. It operates on the principle of electromagnetic induction. When AC voltage is applied to its stator winding, it creates a rotating magnetic field. This rotating magnetic field induces a current in the rotor, which, in turn, generates a magnetic field. The interaction between the two magnetic fields causes the rotor to rotate, driving the mechanical load. Induction motors do not operate in strict synchrony with the supply frequency. The speed of an induction motor can be controlled to some extent using methods like variable frequency drives (VFDs) to change the voltage and frequency applied to the motor.

On the other hand, synchronous generators are used for converting mechanical energy into electrical power while maintaining strict synchronization with the grid frequency. A synchronous generator operates on the principle of maintaining synchrony with the grid frequency. It consists of a stator with windings and a rotor with a magnetic field source (e.g., permanent magnets or field windings). The rotor rotates at a constant speed, synchronous with the grid frequency. As the rotor turns, it induces AC voltage in the stator windings due to the changing magnetic field, generating electrical power. The output frequency of a synchronous generator is fixed and synchronized with the grid frequency, making it less suitable for variable-speed control. However, the voltage and power factor can be controlled.

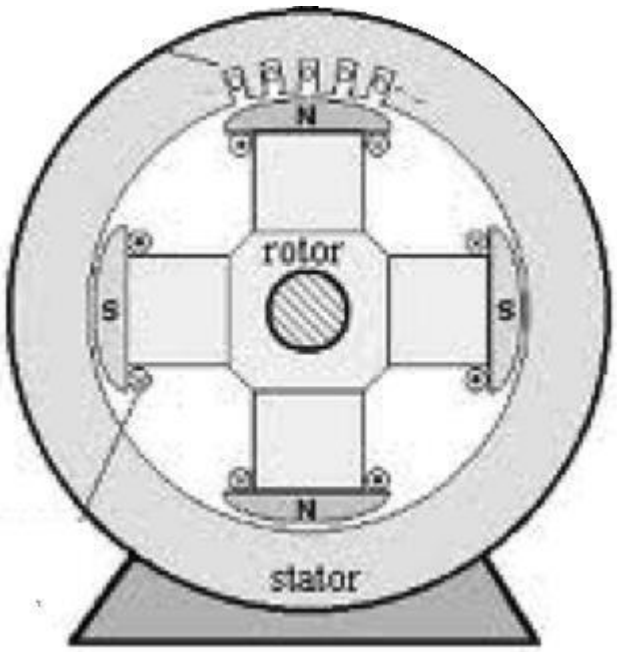


Figure 1 Synchronous Generator

An equivalent circuit of synchronous generator is attached below:

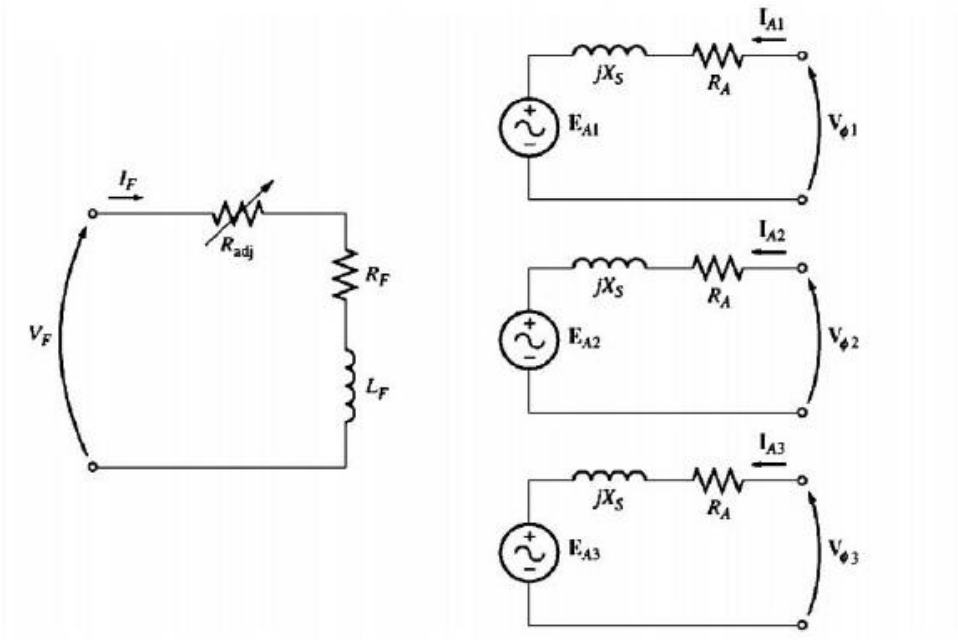


Figure 2: Equivalent Circuit of Synchronous Generator

Synchronous generators are used largely since they have different advantages than other types of generators, like

- They can operate at a wide range of speeds.
- They can be employed to generate AC power at different frequencies.

- They are very effective.
- They are reliable and have a long lifespan.
- This generator can be used to provide backup power in the case of a power outage.

Besides these advantages, there are some complexities.

2.1 Complexity analysis

Constructing a synchronous generator involves several complexities, and understanding these complexities is essential for designing and building efficient and reliable generators. Here are some of the complexities involved in constructing a synchronous generator.

Rotor and Stator Design: our rotor was a squirrel cage rotor which had embedded conductors but for synchronous generator we had to change the shape of our rotor into salient pole for which we had to drill our rotor.

Material Selection: Choosing the right materials for the rotor and stator cores, windings, and insulation is crucial for efficiency and durability. So extensive material testing and research are required to select materials that can withstand high temperatures, magnetic fields, and mechanical stresses.

Synchronization with the Grid: Synchronous generators must operate in synchrony with the grid frequency to avoid instability issues. So, implementing advanced control systems and voltage regulation techniques are needed to maintain synchronization with the grid.

Mechanical Design and Balancing: Mechanical design must ensure that the rotor is balanced to prevent vibrations and mechanical failures. Precision manufacturing and dynamic balancing techniques are essential to achieve optimal mechanical performance.

Maintenance: Synchronous generators require periodic maintenance to ensure long-term reliability. Developing a comprehensive maintenance schedule and monitoring systems to detect issues before they become critical can minimize downtime and repair costs.

Efficiency and Power Factor: It was challenging to maintain an efficiency of around 50% due to the non-uniform air gap in the stator and rotor core. The leakage flux was undesirably high and eddy current loss could not be compensated. We know that the power factor of a synchronous generator varies with field current, so it was difficult to choose the value of rated power factor for design purposes.

Cost Considerations: Due to our lack of experience, materials selection did not consider cost optimization and we needed professional help. As a result, expensive material was used where it was not necessary. For example, our rotor shaft, rotor drilling, mounting of commutator and slip rings etc. cost more than expected.

As constructing a synchronous generator involves a range of complexities related to design, materials, control, cooling, maintenance, and more, it can be said that this problem is a complex engineering problem. However, these complexities can be addressed through advanced engineering practices, modeling and simulation, and the use of modern materials and technologies.

3 Design

3.1 Problem Formulation

To design our synchronous generator from an induction motor, proper calculations and analysis are done. The calculations include total flux in airgap, determination of number of turns per coil in the stator, rated current of the stator coils, gauge of wires and finally the insulation class of the wires. These calculations have helped us achieve PO(b).

3.1.1 Identification of Scope

We wish to convert a 3 phase 4 pole induction motor into a synchronous generator/motor. The changes we must impart construction wise are: -

- Cut-off the old stator wires and replace them with wires of proper AWG value to support full load current.
- Since we must supply DC voltage to the rotor to produce magnetic flux lines, we need to install brush and slip rings on the rotor shaft.
- Our induction motor was a squirrel cage motor, meaning its rotor was smooth cylindrical. We must drill the motor to make it salient pole so that windings can be wrapped.
- Previously, there were no windings in the rotor, now we must wind wires of appropriate rating so that they can handle full load currents.

3.2 Design Method

An alternator consists of two main parts

Stator – The stator is the stationary part of the alternator. It carries the armature winding in which the voltage is generated. The output of the alternator is taken form the stator.

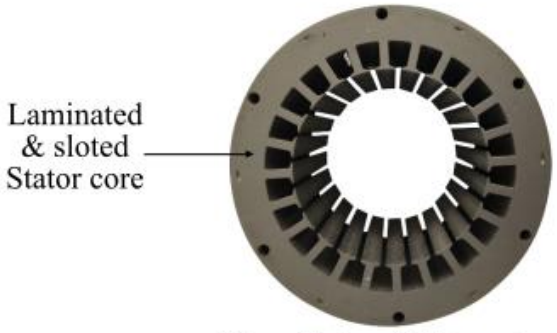


Fig. - Stator of Alternator

Rotor – The rotor is the rotating part of the alternator. The rotor produces the main field flux.

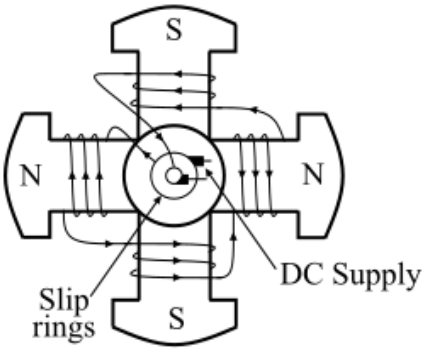


Fig. - Salient Pole Rotor

Stator rewinding, rotor winding and calculation of various parameters of synchronous motor have been done using the knowledge of EEE-205 course which is an achievement of PO(a). first we analyzed how to connect the stator winding and rotor winding for three phase connection. We drew those diagrams. Those are attached below.

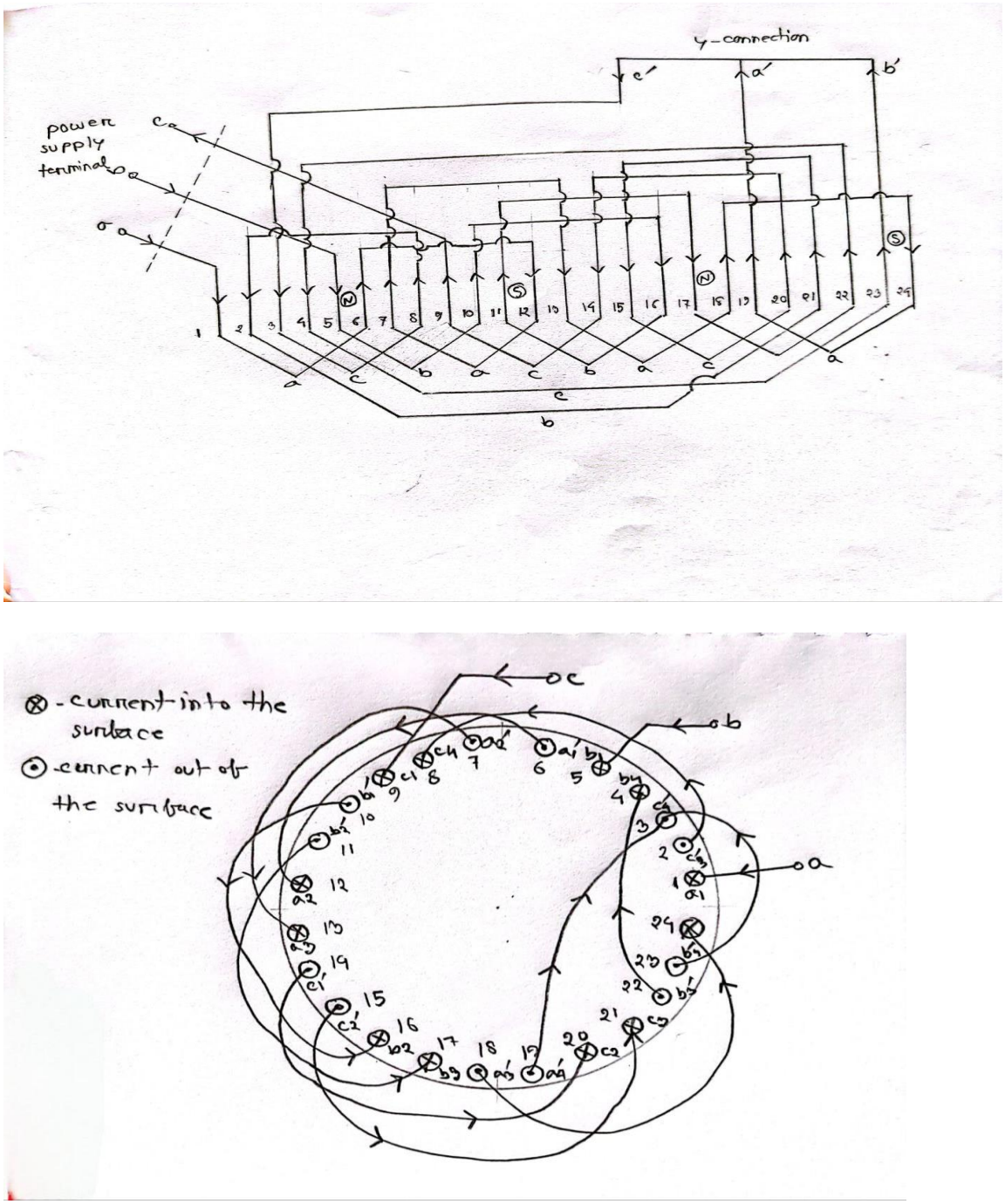
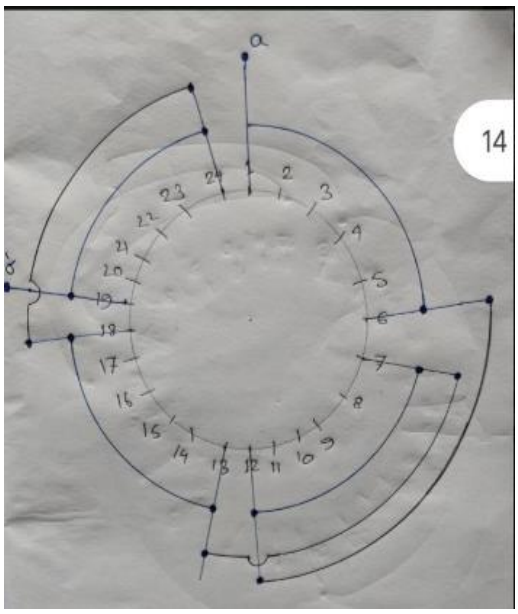
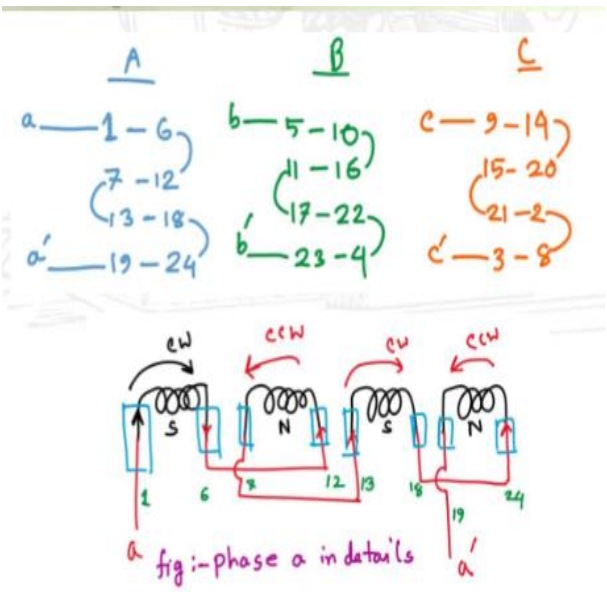


Figure 3 Winding Diagram of 4-Pole 24 Slots 5/6 Pitch Stator





In the next step we calculated different important parameter to take our decision while making the synchronous generator. Form our calculation we selected the perfect wire for our project. The calculation are attached below.

Flux Calculation

The average flux in air gap for 50Hz machine is usually in the range 0.4-0.6 Wb/m^2 and it should never cross 1.8T-2T.
For our design purposes, we assume B_{av}=0.4 Wb/m^2
Core Diameter, D=6cm
Core Length, L=5.5cm
Since the magnetic field lines are incident on the stator at nearly 90 degrees, the total flux is,
$\phi = Bav * \pi DL$ = 0.4*π*6*5.2*10^{-4} W $= 3.9207 * 10^{-3}$ Wb
Flux per pole, $\phi^{'}=\phi/4 = 9.8017 * 10^{-4} Wb$

Turns Calculation

Number of slots, $N_{SL} = \frac{S1}{Pm} = \frac{24}{4*3} = 2$ slots per pole per phase.
Electrical angle per slot, $\gamma = \frac{180*P}{S1} = \frac{180*4}{24} = 30^0$
Distribution factor, $K_d = \frac{\sin(Ns1*\frac{\gamma}{2})}{Ns1*\sin(\frac{\gamma}{2})} = \frac{\sin(2*15)}{2*\sin(15)} = 0.9659$
Our stator is a 5/6 fractional pitch stator. Thus, the mechanical angle that a coil covers, $\theta_m = \frac{5*90}{6} = 75$
Pole pitch, $\rho_p = \frac{360}{P} = \frac{360}{4} = 90^0$ (Mechanical Angle)
Pitch in electrical degrees, $\rho = \frac{\theta m*180}{\rho p} = 150$

Pitch factor, $K_p = \sin\left(\frac{\rho}{2}\right) = \sin\left(\frac{150}{2}\right) = \sin(75) = 0.9659$
The winding factor, $K_W = K_P * K_d = 0.933$
Frequency, f = 50Hz
No. of slots = 24
No of coils (Total) = 24/2 = 12
No of coils per phase = 12/3 = 4
Turns per phase, $t = \frac{E}{\sqrt{2}\pi f \phi' Kw} = \frac{415/\sqrt{3}}{\sqrt{2}\pi * 50 * 9.8017 * 10^{-4} * 0.933} = 1179.4145$
Turns per coil, $N = \frac{1179.4145}{4} = 294.85 \approx 295\ turns$

Rated Current calculation

Rated Line voltage = 415 V
Since, the stator is Y connected, $V_\varphi = \frac{415}{\sqrt{3}} V$
Power rating $P_{3\varphi}$ = 0.4*746 W ≈ 300W
Thus, $P_{1\varphi}$ = 100W
For design purposes, pf = 0.6
The rated line current, $I_{L,\ rated} = \frac{100}{0.6 * \frac{415}{\sqrt{3}}}$ A = 0.6956 A

Wire Size Calculation

Current density for low power motor is 5-10A/mm^2 in our case, δ_s=10A/mm^2
Cross-sectional are of the conductor, A= $\frac{I}{\delta s} = \frac{0.6956}{10} mm^2 = 0.06956$ mm^2
Now we have to follow a standard table for determining our wire gauge.

AWG	Diameter		Cross Section Area			Resistance in Ω		Current in Amps	
	mm	inch	mm^2	$Inch^2$	kcmil	Ω / kft	Ω per km	Chassis Wiring	Power Transmission
0000 (4/0)	11.6840	0.4600	107.2193	0.1662	211.6000	0.049	0.1608	380	302
000 (3/0)	10.4049	0.4096	85.0288	0.1318	167.8064	0.0618	0.2028	328	239
00 (2/0)	9.2658	0.3648	67.4309	0.1045	133.0765	0.0779	0.2557	283	190
0 (1/0)	8.2515	0.3249	53.4751	0.0829	105.5345	0.0983	0.3224	245	150
1	7.3481	0.2893	42.4077	0.0657	83.6927	0.1239	0.4066	211	119
2	6.5437	0.2576	33.6308	0.0521	66.3713	0.1563	0.5127	181	94
3	5.8273	0.2294	26.6705	0.0413	52.6348	0.197	0.6464	158	75
4	5.1894	0.2043	21.1506	0.0328	41.7413	0.2485	0.8152	135	60
5	4.6213	0.1819	16.7732	0.0260	33.1024	0.3133	1.028	118	47
6	4.1154	0.1620	13.3018	0.0206	26.2514	0.3951	1.296	101	37
7	3.6649	0.1443	10.5488	0.0164	20.8183	0.4982	1.634	89	30
8	3.2636	0.1285	8.3656	0.0130	16.5097	0.6282	2.061	73	24
9	2.9064	0.1144	6.6342	0.0103	13.0927	0.7921	2.599	64	19
10	2.5882	0.1019	5.2612	0.0082	10.3830	0.9988	3.277	55	15
11	2.3048	0.0907	4.1723	0.0065	8.2341	1.26	4.132	47	12
12	2.0525	0.0808	3.3088	0.0051	6.5299	1.588	5.211	41	9.3
13	1.8278	0.0720	2.6240	0.0041	5.1785	2.003	6.571	35	7.4
14	1.6277	0.0641	2.0809	0.0032	4.1067	2.525	8.285	32	5.9
15	1.4495	0.0571	1.6502	0.0026	3.2568	3.184	10.448	28	4.7
16	1.2908	0.0508	1.3087	0.0020	2.5827	4.015	13.174	22	3.7
17	1.1495	0.0453	1.0378	0.0016	2.0482	5.063	16.612	19	2.9
18	1.0237	0.0403	0.8230	0.0013	1.6243	6.385	20.948	16	2.3
19	0.9116	0.0359	0.6527	0.0010	1.2881	8.051	26.415	14	1.8
20	0.8118	0.0320	0.5176	0.0008	1.0215	10.152	33.308	11	1.5
21	0.7229	0.0285	0.4105	0.0006	0.8101	12.802	42.001	9	1.2
22	0.6438	0.0253	0.3255	0.0005	0.6424	16.143	52.962	7	0.92
23	0.5733	0.0226	0.2582	0.0004	0.5095	20.356	66.784	4.7	0.729
24	0.5106	0.0201	0.2047	0.0003	0.4040	25.668	84.213	3.5	0.577
25	0.4547	0.0179	0.1624	0.0003	0.3204	32.367	106.19	2.7	0.457
26	0.4049	0.0159	0.1288	0.00025	0.2541	40.814	133.9	2.2	0.361
27	0.3606	0.0142	0.1021	0.0002	0.2015	51.466	168.85	1.7	0.288
28	0.3211	0.0126	0.0810	0.00013	0.1598	64.897	212.92	1.4	0.226
29	0.2859	0.0113	0.0642	0.00010	0.1267	81.833	268.48	1.2	0.182
29	0.2859	0.0113	0.0642	0.00010	0.1267	81.833	268.48	1.2	0.182
30	0.2546	0.0100	0.0509	0.00008	0.1005	103.19	338.55	0.86	0.142
31	0.2268	0.0089	0.0404	0.00006	0.0797	130.12	426.9	0.7	0.113
32	0.2019	0.0080	0.0320	0.00005	0.0632	164.08	538.32	0.53	0.091
33	0.1798	0.0071	0.0254	0.00004	0.0501	206.9	678.8	0.51	0.088
34	0.1601	0.0063	0.0201	0.00003	0.0398	260.9	855.96	0.43	0.072
35	0.1426	0.0056	0.0160	0.00002	0.0315	328.98	1,079.3	0.43	0.072
36	0.1270	0.0050	0.0127	0.00002	0.0250	414.84	1,361	0.33	0.056
37	0.1131	0.0045	0.0100	0.00002	0.0198	523.1	1,716.2	0.33	0.056
38	0.1007	0.0040	0.0080	0.00001	0.0157	659.62	2,164.1	0.27	0.044
39	0.0897	0.0035	0.0063	0.00001	0.0125	831.77	2,728.9	0.26	0.043
40	0.0799	0.0031	0.0050	0.00001	0.0099	1,048.8	3,441.1	0.21	0.035

Figure 4 AWG chart

From the table we can see that our wire should be of **29 AWG** value.

Rotor Wire Gauge: 31 AWG
No of turns: 1800

4 Implementation

4.1 Description

Step by Step procedure

Step-1: Purchasing an Induction Motor of specific rating



Figure 5 Induction motor

A 3 phase 4 pole motor with power rating of 1HP, rated voltage of 440V, 50Hz was bought for our purpose. The nameplate of purchased motor is shown below:

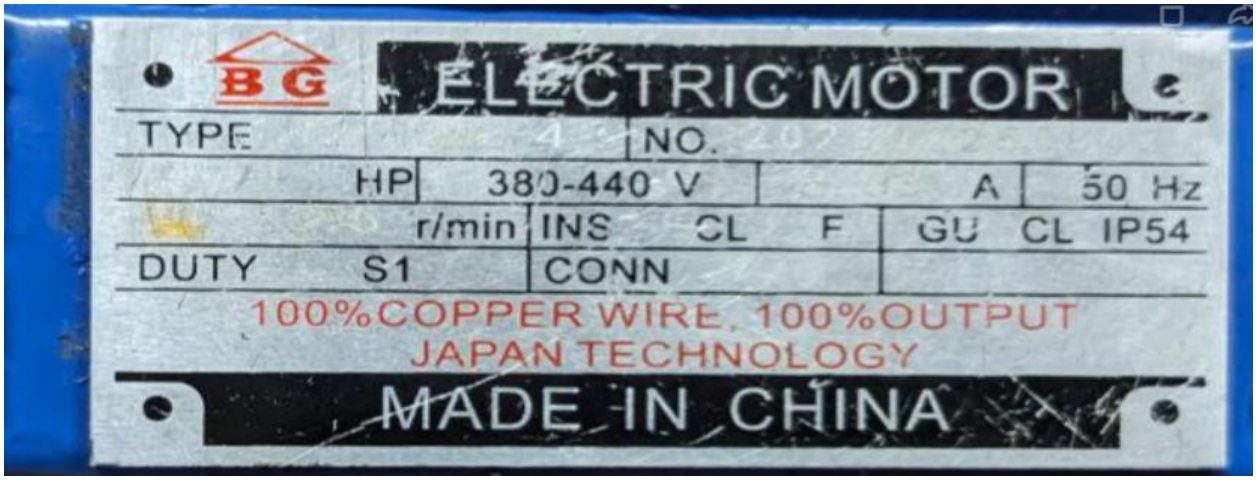


Figure 6: Motor nameplate

Step-2: Disassembling the motor

The induction motor was disassembled by unscrewing all the nuts, removing the cooling fan from shaft, removing the chassis of the motor. The shaft forced out by hammering. After all this, the motor looked something like the following:

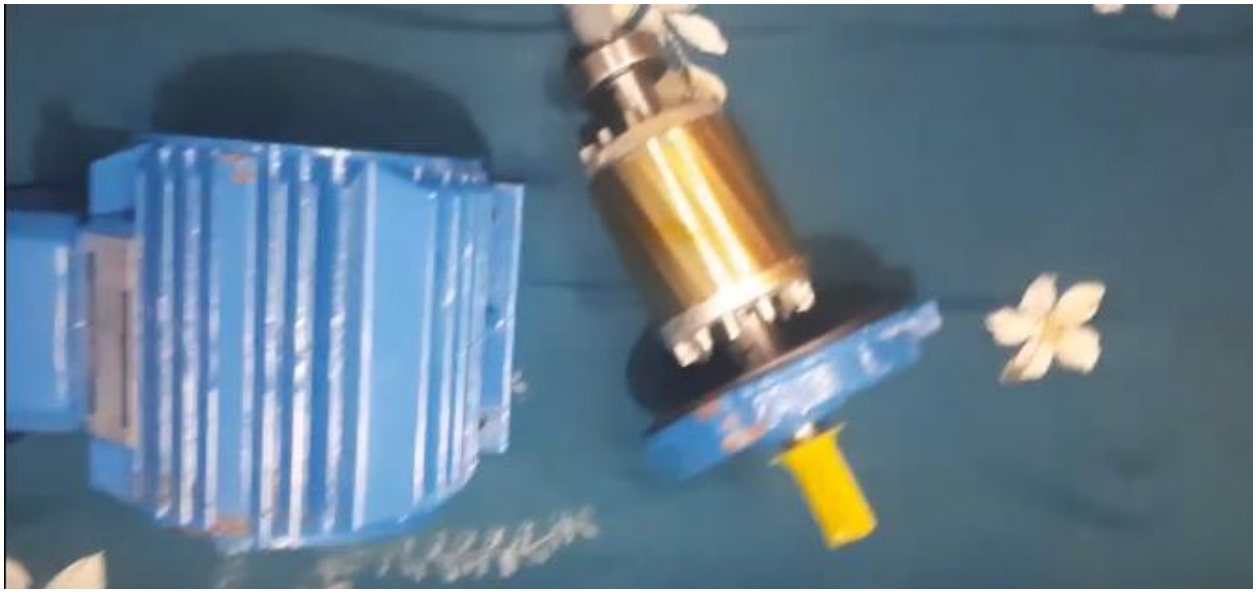


Figure 7: After motor dissembling

Step-3:

The old armature coils were cut-off and old insulation was removed. New insulation paper was installed in the freed slots.





Figure 8: Stator winding dissembling



Figure 9 Stator after installation of insulation

Step-4: Preparation of stator coils

Cu wires of 29 AWG value wrapped around a forma to prepare our coils. Each coil consisted of nearly 300 turns.



Figure 10 Stator Coil

Step-5: Rewinding of stator coils

The stator coils were re-winded following proper winding diagram. The details of winding diagram are provided in chapter 3.2- (Design Method). Insulation was placed between coils to prevent short circuit current flow between coils.





Figure 11 Stator after rewinding

Step-6 (Rotor Drilling)

Initially our rotor was cylindrical in shape and contained no windings rather only conductors were embedded. We had to drill our rotor to convert it into salient pole. Four 13mm drills were conducted



Figure 12 Rotor post drilling

to achieve this trait.

Step-7: (Rotor winding)

Two coils each of 1800 turns were inserted to effectively create a 4 pole device.



Figure 13 Rotor WInding

Step-8: Final assembly with commentator, slip rings and brushes mounted on the shaft

The chassis was put back together and Commutator, Brush and slip rings were mounted on rotor shaft to complete our setup



Figure 14 Final assembly

4.2 Experiment and Data Collection

During the measurement of the resultant circuit of the synchronous generator, we find the three parameters that are very important to study the complete performance of the real-world synchronous generator. These three parameters are described here.

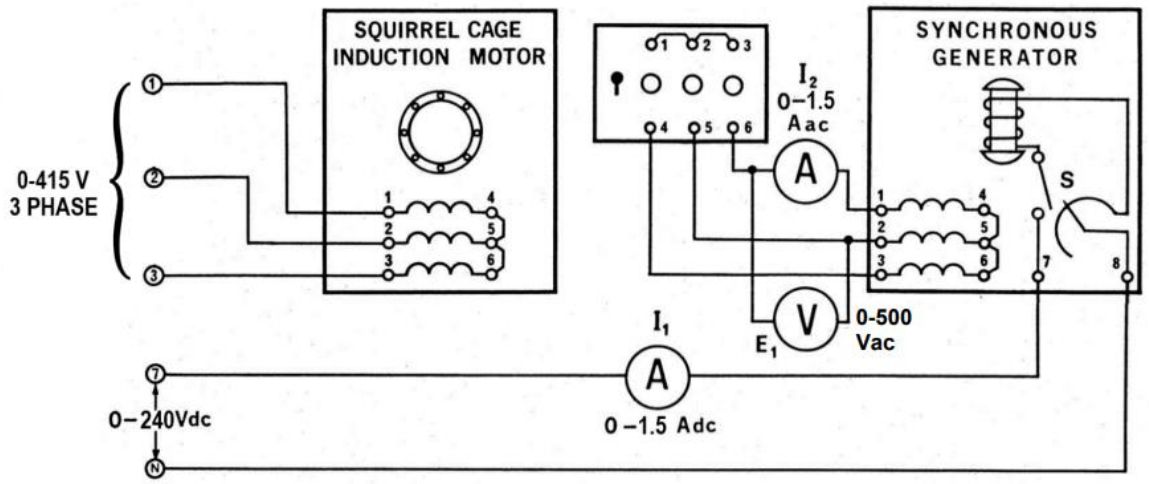
- The armature resistance.
- The relation between field current and internally generated voltage
- Synchronous reactance X_s

To find these three parameters we will perform some tests on the synchronous generator that are described here in detail.

Open Circuit Test of the Synchronous Generator

- To accomplish this test the generator should move at its rated speed, no-load should be at its terminals, and the value of the field winding current should be 0.
- After following these steps, we will vary the field current step by step and calculate the terminal voltage at each step.
- As there is no load at the output terminals of the generator, there will be zero armature current.
- Hence the internally generated voltage E_A will equal the phase voltage $V_ø$.
- By using this data, we can construct the curve among the internally generated voltage that equals the terminal voltage and the field current.
- This graph between E_A and the field current is known as the **open-circuit characteristic** (OCC) of the synchronous generator.
- By this graphical representation, we can find the value of internal generated voltage E_A any value of the field current.

Possible experimental setup diagram is given below:



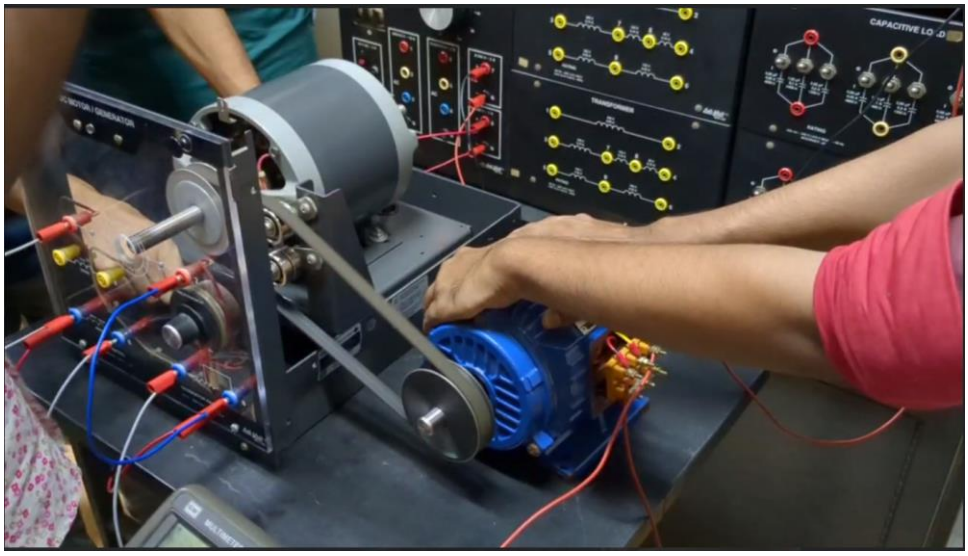


Figure 15 OCC test

Data Table (OCC test):

Field current	Output Voltage (V)
0	9.87
0.1	62
0.2	148
0.3	212
0.4	257
0.5	288
0.6	307
0.7	321
0.8	333
0.9	341
1.0	356

Short circuit test of Synchronous Generator

- To do a short circuit test first of set the value of field current at 0 and connect the output terminals of the generator by the ammeter.
- After that we have to find the value of the armature current (I_A) by changing the field current.
- The graph among the field current and the armature current known as the short-circuit characteristic (SCC).
- The graph becomes linear

4.3 Data Analysis

The open-circuit (OC) test of a synchronous generator is a test performed to determine the open-circuit voltage of the generator. The test is performed by connecting the generator to a load that draws no current and then measuring the generator's terminal voltage. From the collected data we plotted the OCC curve in the MATLAB and the corresponding graph is attached here.

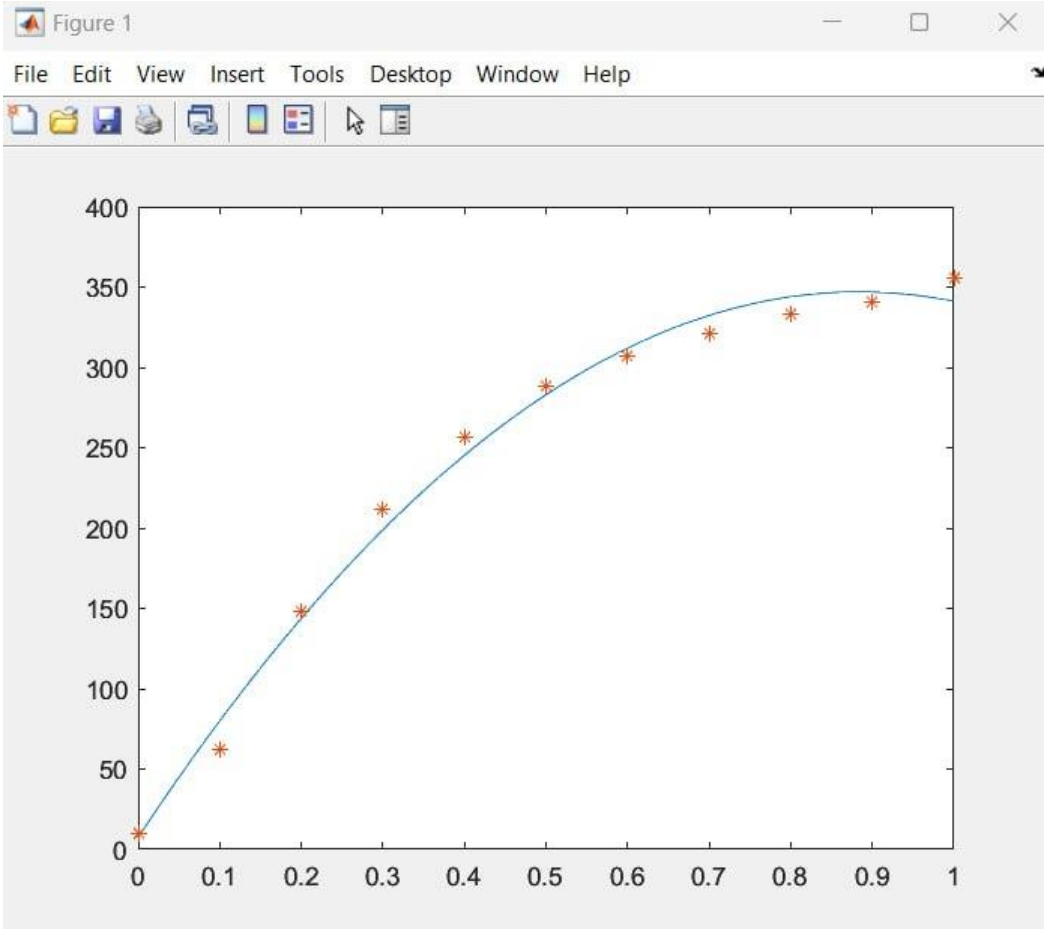


Figure 16: OCC curve

MATLAB code:

```
clc;
clearvars;
close all;
x=[0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1];
y=[9.87 62 148 212 257 288 307 321 333 341 356];
n=length(x);
A=zeros(n,3);
A(:,1)=1;
A(:,2)=x;
A(:,3)=x.^2;
Y=y';
s=pinv(A'*A)*A'*Y;
x_new=0:0.01:1;
y_new=s(1,1)+s(2,1).*x_new+s(3,1).*x_new.*x_new;
plot(x_new,y_new,x,y,'*');
```

From the graph, we can see that the OCC curve remains linear up to certain excitation. As the field excitation is increased, the generator's terminal voltage also increases. This is because increasing the field excitation increases the magnetic field strength of the generator, which in turn increases the induced voltage in the generator's windings. But after about 0.6 A non-linearity is seen. This is because the generator's magnetic circuit is not linear. As the field excitation is increased, the magnetic saturation of the generator's core increases, which causes the relationship between If and Va to become non-linear.

4.4 YouTube Link

We have captured some of our work as video clip. All of them are compiled in a single video. The video link is given below:

https://www.youtube.com/watch?v=5CahcHJUpLI

5 Design Analysis and Evaluation

5.1 Design Considerations

5.1.1 Considerations to public health and safety

Electricity Security:

Throughout the project, strict adherence to electrical safety regulations has been of the utmost importance. All people involved must utilize protective equipment, handle insulated instruments properly, and properly ground themselves. To guarantee that electrical components are operating properly and to guard against any potential electrical hazards, regular safety inspections have been carried out.

Material Choice:
Copper wire has been specifically chosen for the wiring in the generator because of its outstanding electrical conductivity and non-toxic nature. This decision reduces the possibility of dangerous material interactions and guarantees secure electrical connections.

Risk Assessment:
Before the project started, a thorough risk analysis was done. This assessment identified, ranked, and prioritized potential conversion-related risks and hazards.
A thorough risk management plan was created using the results of the risk assessment to make sure that dangers are properly handled and reduced.

5.1.2 Considerations to the environment

Sustainability and the selection of Materials:
Emphasis has been focused on selecting materials with the least negative influence on the environment. Sustainable and environmentally friendly materials have traditionally been selected, such as recyclable metals and non-toxic insulators.

Resource Efficiency:
In order to reduce waste production, we have optimized the usage of materials with an emphasis on resource efficiency. Materials that can be recycled or properly disposed of without harming the environment.

Energy Efficiency:
The synchronous generator has been built with excellent energy efficiency in mind. This aids in environmental preservation by lowering greenhouse gas emissions as well as energy usage.

Commitment to Environmental Sustainability:
Beyond the project's conclusion, our project team is dedicated to environmental sustainability. To lessen our overall environmental imprint, we are looking into possibilities for integrating renewable energy and other eco-friendly measures.

5.1.3 Considerations of cultural and social needs

In our pursuit to convert an induction motor into a synchronous generator, it is essential to acknowledge and address the cultural and societal aspects that may be influenced by this project. Cultural and societal considerations play a vital role in ensuring that our work is respectful, inclusive, and aligns with the needs and values of the communities involved. This report outlines the various steps and considerations taken during the conversion process to accommodate cultural and societal needs effectively.

5.2 Limitations of Tools

To convert our rotor from cylindrical to salient pole we had to change its shape, that is we had to conduct four 13 mm drills on it, but we did not have the proper tools or safety measures for this task. We also had to conduct CNC cutting on our rotor which was also not possible for us. we needed to mount a commutator, brush and slip ring setup on the rotor shaft but we did not have the proper instruments for this task

5.3 Impact Assessment

5.3.1. Assessment of Societal and Cultural Issues

We assessed the possibilities of our project if executed on a larger industrial level. Synchronous generators can be integrated with the renewable energy production system for better stability and performance. This is a huge potential sector for rural and suburban areas with power shortage. This energy production system will need adequate and skilled manpower. So, the career-oriented dynamics of that area would revolve around the power plant and sectors connected to it.
In the proposed case we would have to ensure that the operation and maintenance of our generator and corresponding systems do not hinder the daily life and workflow of the society around it.

Government and non-governmental organizations need to take initiatives to integrate synchronous generators in the power production system and solve the social and cultural issues related to it.

5.3.2 Assessment of Health and Safety Issues

We ensured proper grounding and insulation measures to prevent electrical shock hazards. We did not use any kind of harmful chemical in the process of designing the generator that could effect of health severely. For compliance and future reference, we kept records of safety inspections, maintenance tasks, and safety occurrences. We conducted regular risk analyses to find and eliminate the generator's new or evolving safety risks. We took extra careful measures while working with the moving parts of the generator.

In short, we ensured that every member of our group has proper knowledge about the operation of the generator and necessary steps in case of any hazardous situation.

5.3.3 Assessment of Legal Issues

We tried to ensure that the amount of emission and waste stays below the limit instructed by the existing environmental laws in effect. We stayed informed about potential changes in regulations that may potentially affect the operation, emissions or safety standards of our synchronous generators.

As our generator was not designed for commercial purposes. We did the project to demonstrate our engineering expertise based on theoretical knowledge acquired in our ongoing semester. Therefore, we did not have to acquire licenses or permits. But we were fully aware of the rules and assessed possible scenarios in which our project would be industrially used.

5.4 Ethical Issued

The main ethical concern was upholding the ideals of independence and academic integrity. Every step of the task including rotor rewinding, stator winding, motor assembly and disassembly, and insulation paper installation was completed by the group members without the use of outside resources. Only professional assistance was taken in tasks that were not feasible for the students to do or we didn’t have the required instruments to do so. It was crucial to uphold moral standards in order to avoid any kind of plagiarism or unlawful collaboration. All the members contributed equally to complete the project. This strategy guaranteed that the project's results remained an authentic representation of each person's work and knowledge.

6 Reflection on Individual and Teamwork

6.1 Mode of Teamwork:

1. Clear Project Goals and Objectives: We defined the project's specific objectives, purpose and scopes to all our project members. We ensured that all the team members understood and aligned with these goals. Each team member has had clearly defined role and responsibilities.

2. Communication and Collaboration: We established open and effective channels of communication among team members and held regular team meetings to discuss progress, issues, and updates. We made sure that everyone was kept informed of the latest developments and that there were scopes for open communication.

3.Budget and Resource Management: We developed a detailed budget for the project, including expenses for equipment, personnel, professional and all other sectors. We continuously monitored expenses and allocated resources efficiently.

4. Task and Deadline Management: We worked according to the project timeline with deadlines for different project milestones. We encouraged effective time management by setting priorities. We appointed a few members from our group responsible for overseeing the project's progress, coordinating activities, and making decisions.

5.Feedback and Peer Review: We encouraged peer review of each team member's contributions and provided constructive feedback to help each other improve.

6. Flexibility and Adaptability: We were open to adjusting the project plan if necessary due to unforeseen circumstances. We were flexible and adapted to changes as a team.

7. Post-Project Evaluation: We conduct a comprehensive evaluation of the project upon completion. We identified any unresolved issues, solved them accordingly with the proper documentation and reporting. We also identified the lessons learned and best practices for future power projects.

6.2 Diversity Statement of Team

The habituation of our group members was very diverse since many of our group members came from distant places for project work. Each of our members have their own unique quality and area of specialization for example some were very skilled in video editing , some had good capture on circuit construction and wiring , some had a good grasp over the relevant theories which helped us in our design procedure. Some of our members conducted simulation for dry run so that no problem occurred when it will be tested in the laboratory.

6.3 Logbook of Project & Contribution

Date	Milestone achieved	Individual Contribution	Team Role	Comments
5 July, 2023	Buying an Induction Motor	2006133,2006153,2006156		
10 July, 2023	Dismantling the Induction Motor	2006134,2006152,2006131 2006135,2006151,2006132		
20 July, 2023	Cutting the Stator Winding of the Motor	2006150,2006151,2006136, 2006155,2006153,2006135, 2006131		
	Calculating the Winding of the Rotor and Stator Needed for Synchronizing	2006134,2006153,2006152, 2006135,2006136,2006155		
5 August, 2023	Buying wire	2006150,2006131,2006154		
9 August, 2023	Rewinding the Rotor		Everyone	
9 August, 2023	Rewinding the Stator		Everyone	
21 August, 2023	Properly Insulating the Windings	2006136,2006133,2006132, 2006135,206134,2006156		
24 August, 2023	Setting Up Slip Ring and Brushes			Expert help
5 September, 2023	Assembling the generator	2006150,2006152,2006155, 2006154,2006132,2006151		
10 September, 2023	OCC Test	2006133,2006134,2006136, 2006153,2006150,2006131, 2006155,2006152,2006156		

7 Communication

Giving and Receiving Clear Instructions:

The smooth completion of numerous tasks was made possible by our team's communication throughout the project. We got clear directions from our course teacher. To accomplish the appropriate winding configuration, clear directions were given during the rewinding process. Exact guidelines were also followed during the rotor drilling procedure to provide mechanical stability. We also actively participated in getting comments from our seniors, mentors, peers, and professors. This recurrent feedback loop helped us improve our procedures and showed that we were willing to consider helpful criticism.

Comprehending and Writing Effective Reports:

The capacity to comprehend the complex engineering procedures involved in converting the motor into a synchronous generator and then transform this comprehension into a clear and coherent written language is one of the essential components of good communication. Each stage of the conversion process is thoroughly described in our project report. The rewinding methods used, the justification for rotor drilling, and the many adjustments necessary to ensure synchronous functioning have all been covered. The report's logical organization makes it possible for readers with different levels of technical expertise to follow the transformation process with ease.

Design Documentation:

The induction motor's adjustments for its transformation into a synchronous generator must be communicated through design documentation, which is crucial in this regard. The final synchronous generator configuration, rotor adjustments, and detailed diagrams, schematics, and technical drawings are all shown in the paper. These visual aids not only improve the readability of the report but also give a visual picture of the engineering choices and changes made.

Effective Presentations:

We have created an interesting and instructive presentation as part of our project's communication efforts to successfully communicate the important elements of our project. The technical background, difficulties encountered, approach used, and results are all covered in this presentation. To help our readers better understand the conversion process and its importance, we have included visual aids like animations, graphs, and graphics to break down difficult ideas.

7.1 Executive Summary

The main goal of our project is to transform typical and inexpensive 3-phase 4-pole induction motors into adaptable synchronous motors/generators. Synchronous motors are perfect for many applications because they maintain a constant speed unlike induction motors, which often work at varying rates. To take advantage of synchronous machines' power generation, constant speed applications, and power factor correction capabilities, we want to develop an easy-to-use conversion technique. When compared to comparably sized induction motors,

synchronous motors are more efficient and have no slip, making them a valued asset. The building and conversion process are examined in this study, with a focus on how easily synchronous generators may be modified to meet different electrical requirements.

7.2 User Manual

The voltages and currents of the three phases of a synchronous generator are 120° apart in angle, but otherwise, the three phases are identical. These three phases can be either Y or Δ connected as shown in below figures.

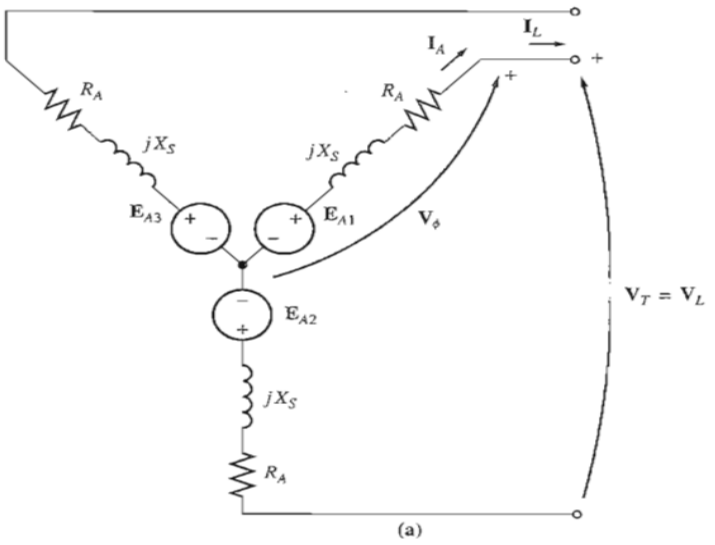


Figure 17 Y connection

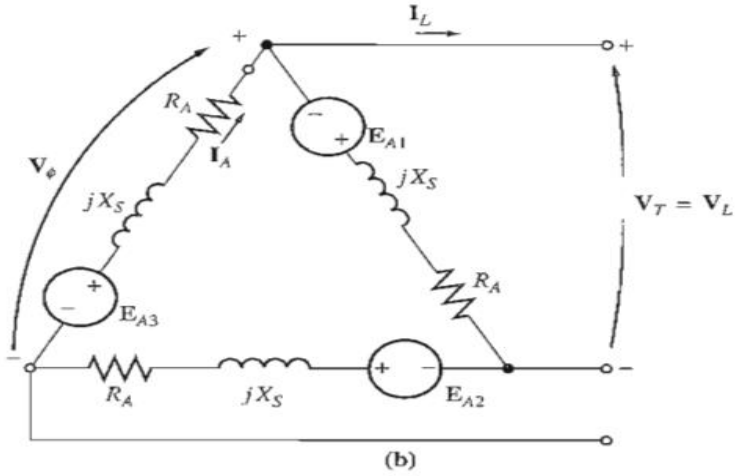


Figure 18 Delta connection

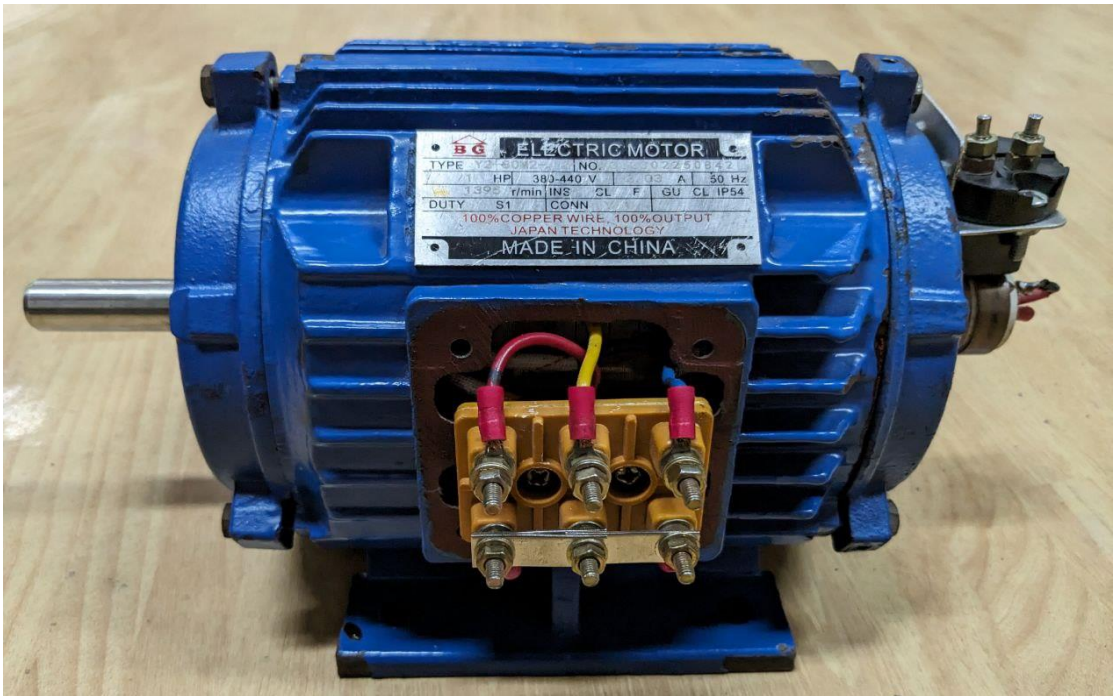


Figure 19 Armature Terminals

From the above figure we can see that the stator terminals are Y-connected since three terminals are shorted via bar and the other three are open. The generated voltage will be found from the above terminals that are colored coded with blue yellow and red. The three leads represent the three phases of our generated voltage.

If the user wishes to connect the stator winding in the delta configuration, then they have to remove the bar that shorts the three terminals and creates the neutral point. This can be done by removing the three nuts afterwards a1 should be connected to b2, b1 should be connected to c2 and c1 should be connected to a2.

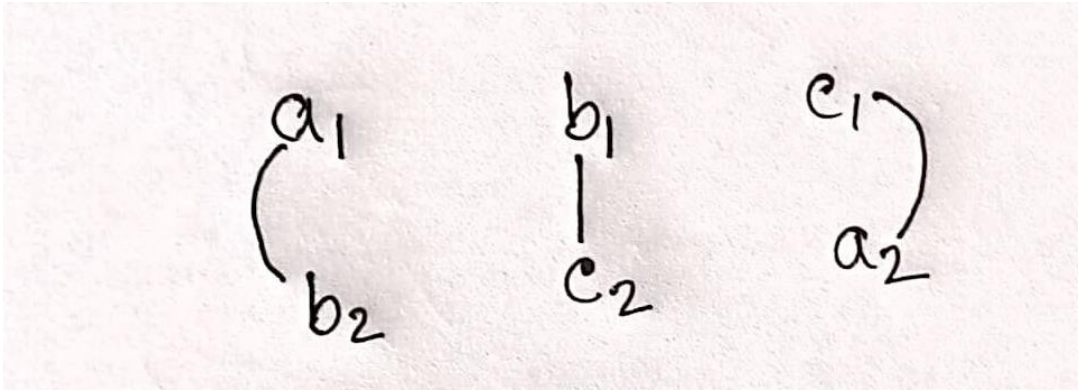


Figure 20 Delta connection diagram

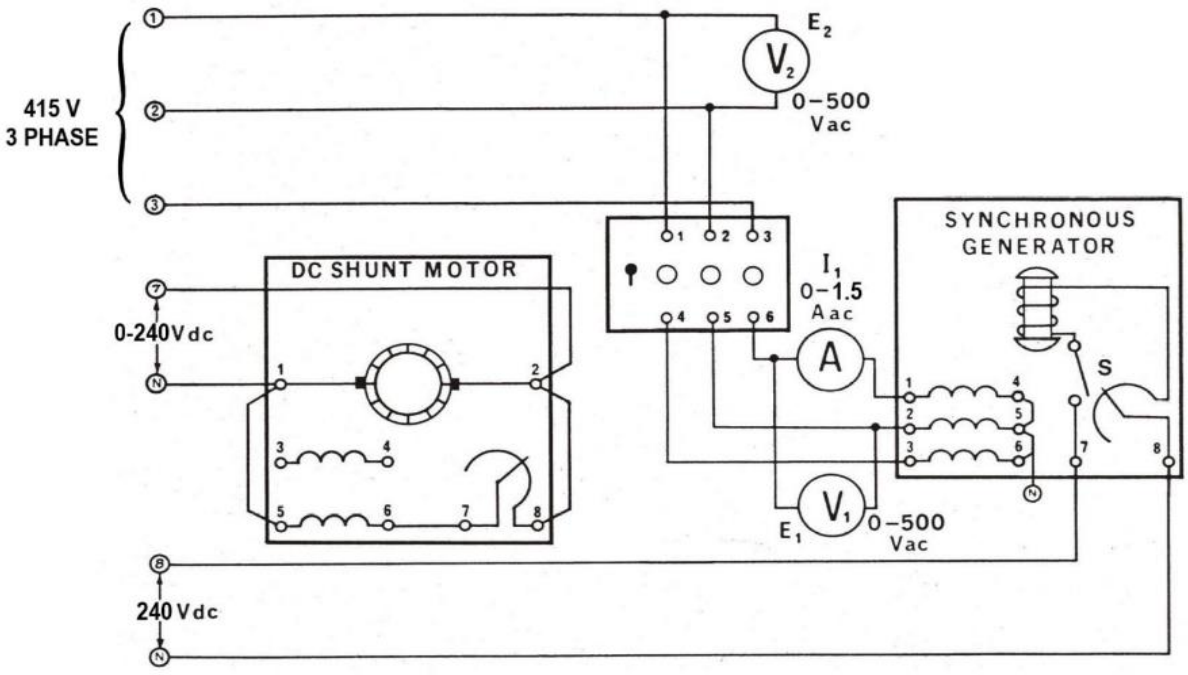


Figure 21 Generator Synchronization

- User can use DC shunt motor as the prime mover to start the synchronous generator. In that case, at first, he has to connect the terminals of the dc shunt motor to a variable (0-240) dc supply.
- The rotor part of the synchronous generator has to be connected to a fixed dc supply(240V)
- For Y configuration lower terminal of the stator are shorted via bar and other three terminals (red, blue, yellow) will generate voltage.
- If user wills to connect the generator to a infinite bus then the stator-colored terminals has to be connected to the 3 phase supply(415V) through a synchronous module. By using the synchronous module user has to ensure that the supply and stator terminals has same rms voltage, same phase sequence. The generator frequency will have be slightly higher than the bus frequency for the generator to contribute in the system.

8 Project Management and Cost Analysis

First, we bought an induction motor on 5 July 2023. Then we cut the winding of the motor on 20/07/2023. After that, we placed the insulator in the slot which took about 3 days. Then we rewinded the coil which took about 9 days. After placing the slot, winding and burnishing took about 10 days. Drilling the rotor, and placing the insulator and coil took about 10 days. We spent about 3 days connecting the brush and slip ring. To connect the coil in the rotor 3 days were needed.

8.1 Cost of Materials

Name of the equipment	Cost (Taka)
Induction Motor	5400
29-gauge stator coil, insulating paper, burnish	1900
31-gauge rotor coil, slip ring, brush and pulley	2360
Commutator	350
Total	10010

8.2 Cost of other activities

Activity	Cost (Taka)
Rotor drilling	1500
Attaching the pulley, slip rings and brush to the rotor	600
Transportation	500
Total	2600

Overall cost=12610 Taka

8.3 Timeline of Project Implementation

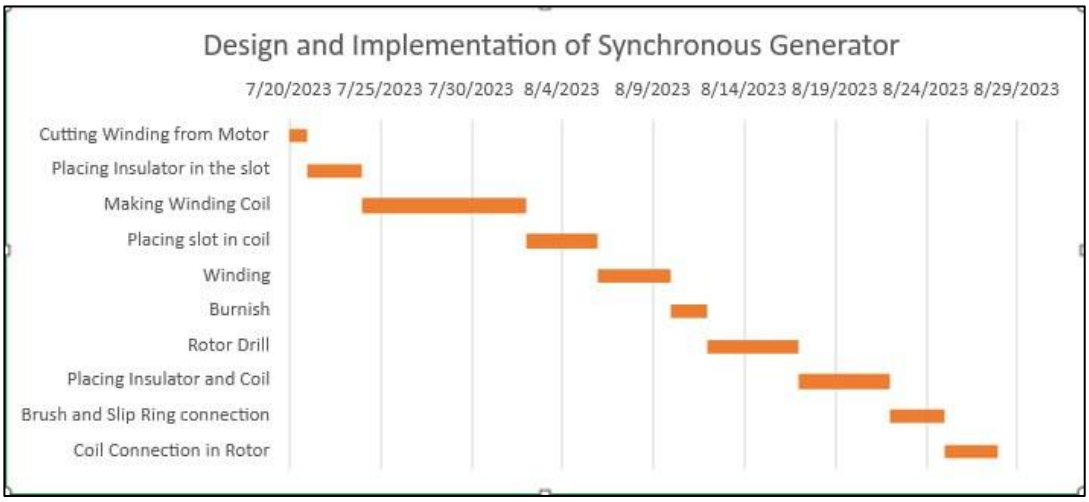


Figure 22 Gantt Chart

Tasks	Starting Date	Ending Date	Total Days
Cutting Winding from Motor	7/20/2023	20/07/2023	1
Placing Insulator in the slot	7/21/2023	23/07/2023	3
Making Winding Coil	7/24/2023	1/8/2023	9
Placing slot in coil	8/2/2023	5/8/2023	4
Winding	8/6/2023	9/8/2023	4
Burnish	8/10/2023	11/8/2023	2
Rotor Drill	8/12/2023	16/8/2023	5
Placing Insulator and Coil	8/17/2023	21/8/2023	5
Brush and Slip Ring connection	8/22/2023	24/8/2023	3
Coil Connection in Rotor	8/25/2023	27/8/2023	3

Figure 23 Time line

We successfully made economic decisions and managed the project finance and thus achieved PO(k).

9 Future Work

Synchronous generators can provide stable and synchronized AC power for which they are used to produce power in various sectors, especially in large scale electricity grids.

But with the growing advancements of energy generation technology, especially the dependency shift towards renewable energy, the implementation of synchronous generator must also evolve. Some of the ways for synchronous generators to evolve are stated below.

- **Renewable Integration:** Synchronous generators are less flexible to fluctuations which renewable energy supply frequently observe. So, synchronous generators that could adapt to renewable power sources might be a future development way for synchronous generators.

- **Energy Storage:** Energy storage is an important aspect in power generation of modern times. People like to store electric energy for future use. And this trend might even expand to large scale power grids in future. Synchronous generator falls short in this case. If synchronous generators could be made more advanced in energy storage, they might hold up to the modern power generation demand.
- **Grid Modernization:** With the advancement of technology, power grids are becoming smarter and more interconnected. These smart power grids require more flexible and adaptable power generating sources. Synchronous generators can be modernized in flexibility and adaptability to cope up with these smart grids in future.

Though synchronous generators play a significant part in large scale power generation, there is still scope for development in future, especially in the case of renewable sources and flexibility.

10 References

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